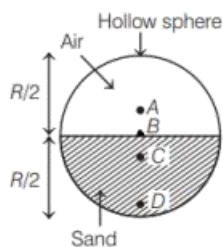


shown in figure?



- (a) A (b) B (c) C (d) D

Ans: (c) C

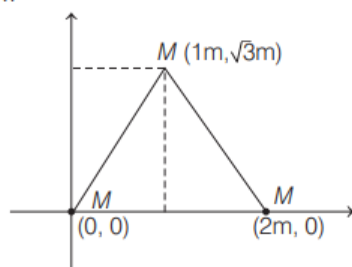
Centre of mass of a system lies towards the part of the system, having bigger mass. In the above diagram, lower part is heavier, hence CM of the system lies below the horizontal diameter.

5. Three identical spheres each of mass M are placed at the corners of an equilateral triangle of side $2m$. Taking one of the corners as the origin, the position vector of the centre of mass is

- (a) $\sqrt{3}(\hat{i} - \hat{j})$ (b) $\frac{\hat{i}}{\sqrt{3}} + \hat{j}$ (c) $\frac{\hat{i} + \hat{j}}{3}$ (d) $(\hat{i} + \frac{\hat{j}}{\sqrt{3}})$

- (d) $(\hat{i} + \frac{\hat{j}}{\sqrt{3}})$

The given system of spheres is as shown below



The x and y -coordinates of centre of mass is

$$x = \frac{\sum m_i x_i}{\sum m_i} = \frac{M \times 0 + M \times 1 + M \times 2}{M + M + M} = 1$$

$$y = \frac{\sum m_i y_i}{\sum m_i} = \frac{M \times 0 + M (\sqrt{3}) + M \times 0}{M + M + M}$$

$$\Rightarrow y = \frac{\sqrt{3} M}{3 M} = \frac{1}{\sqrt{3}}$$

So, position vector of the centre of mass is

$$\left(\hat{i} + \frac{\hat{j}}{\sqrt{3}} \right).$$

6. Newton's second law for rotational motion of a system of a particle can be represented as (L for a system of particles)

- (a) $\frac{dp}{dt} = \tau_{\text{ext}}$ (b) $\frac{dL}{dt} = \tau_{\text{int}}$ (c) $\frac{dL}{dt} = \tau_{\text{ext}}$ (d) $\frac{dL}{dt} = \tau_{\text{ext}} + \tau_{\text{int}}$

Ans: (c) $\frac{dL}{dt} = \tau_{\text{ext}}$

7. A disc of mass M and radius R is rotating about one of its diameters. The value of radius of gyration for the disc is



- (a) $R/4$ (b) $R/2$ (c) $R/6$ (d) $R/8$

Ans: (b) $R/2$

$$I_{\text{disc}} \text{ about the axis along its diameter} \\ = \frac{MR^2}{4} \quad \dots(i)$$

$$\text{Using radius of gyration, } I = Mk^2 \quad \dots(ii)$$

$$\text{Comparing Eqs. (i) and (ii), we get } k = \frac{R}{2}.$$

8. By keeping the moment of inertia of a body constant, if we double the time period, then the angular momentum of the body

- (a) remains constant (b) becomes half (c) doubles (d) quadruples

Ans: (b) becomes half

We know that, angular momentum of the body is given by

$$L = I\omega \text{ or } L = I \times \frac{2\pi}{T} \text{ or } L \propto \frac{1}{T} \Rightarrow \frac{L_1}{L_2} = \frac{T_2}{T_1} \\ \Rightarrow \frac{L}{L_2} = \frac{2T}{T} \quad (\text{as, } T_2 = 2T \text{ and } L_1 = L)$$

So, $L_2 = \frac{L}{2}$. Thus, on doubling the time period, angular momentum of body becomes half.

9. Two bodies have their moments of inertia I and $2I$ respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momenta will be in the ratio

- (a) 1:2 (b) $\sqrt{2}:1$ (c) 2:1 (d) 1: $\sqrt{2}$

Ans: (d) 1: $\sqrt{2}$

Rotational kinetic energy remains same.

$$\text{i.e. } \frac{1}{2} I_1 \omega_1^2 = \frac{1}{2} I_2 \omega_2^2$$

$$\text{or } \frac{1}{2I_1} (I_1 \omega_1)^2 = \frac{1}{2I_2} (I_2 \omega_2)^2$$

$$\Rightarrow \frac{L_1^2}{I_1} = \frac{L_2^2}{I_2} \text{ or } \frac{L_1}{L_2} = \sqrt{\frac{I_1}{I_2}}$$

$$\text{But } I_1 = I, I_2 = 2I$$

$$\Rightarrow \frac{L_1}{L_2} = \sqrt{\frac{I}{2I}} = \frac{1}{\sqrt{2}} \Rightarrow L_1 : L_2 = 1 : \sqrt{2}$$

10. A body having a moment of inertia about its axis of rotation equal to 3 kg m^{-2} is rotating with angular velocity of 3 rad s^{-1} . Kinetic energy of this rotating body is same as that of a body of mass 27 kg moving with a velocity v . The value of v is

- (a) 1 ms^{-1} (b) 0.5 ms^{-1} (c) 2 ms^{-1} (d) 1.5 ms^{-1}

Ans: (a) 1 ms^{-1}

We know that, kinetic energy,

$$K = \frac{1}{2}mv^2 = \frac{1}{2}I\omega^2$$

Given, $m = 27 \text{ kg}$ (mass of the body),

$\omega = 3 \text{ rads}^{-1}$ (angular velocity)

and $I = 3 \text{ kg-m}^2$ (moment of inertia)

$$\Rightarrow mv^2 = I\omega^2 \Rightarrow v^2 = \frac{I\omega^2}{m}$$

$$v^2 = \frac{3 \times 3^2}{27} \Rightarrow v^2 = \frac{27}{27} = 1$$

$$\Rightarrow v = \sqrt{1} = 1 \text{ ms}^{-1}$$

11. An isolated particle of mass m is moving in a horizontal plane (x - y) along the x axis at a certain height above the ground. It suddenly explodes into two fragments of masses $m/4$ and $3m/4$. An instant later, the smaller fragment is at $y = +15 \text{ cm}$. The larger fragment at this instant is at

(a) $y = -5 \text{ cm}$ (b) $y = +20 \text{ cm}$ (c) $y = +5 \text{ cm}$ (d) $y = -20 \text{ cm}$

Before explosion, particle was moving along x -axis, i.e., it has no y -componet of velocity. Therefore the centre of mass will not move in y -direction. or we can say $y_{CM} = 0$

$$\text{now, } y_{CM} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$\text{Therefore, } 0 = \frac{\left(\frac{m}{4}\right)(+15) + (3m/4)(y)}{\left(\frac{m}{4} + 3m/4\right)}$$

$$\text{or } y = -5 \text{ cm}$$